

How Long Is the Graph of x^n ?

Logarithmic Corner Defects in Arc Length

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Abstract—The graph of $y = x^n$ on $[0, 1]$ has arc length $L(n) \rightarrow 2$ as $n \rightarrow \infty$, because the curve collapses onto the two legs of a right-angled corner. We determine the rate of this convergence. Comparing Euclidean with taxicab length and evaluating a single Beta integral, we derive an exact, rapidly convergent representation of $L(n)$ in terms of the Gamma function. From it we obtain $2 - L(n) = (\ln n + 1 - \ln 2)/(n-1) + O(\ln^2 n/n^2)$, the sharper approximation $L(n) = 1 + (2/(en))^{1/(n-1)} + \pi^2/(12n^2) + O(\ln n/n^3)$, and elementary two-sided bounds valid for all $n \geq 3$. The logarithm arises because the curve approaches the vertical leg of the corner only logarithmically. For a corner of arbitrary turning angle θ , the coefficient of $\ln n/n$ becomes $2 \sin^2(\theta/2)$, and we find the next-order constant in closed form. By contrast, the perimeter of the superellipse $|x|^p + |y|^p = 1$ satisfies $P(p) = 8 - 4K/p + O(p^{-2})$ with $K = 2\sqrt{2} \ln(1 + \sqrt{2}) - 2 \ln 2$, with no logarithm, because the superellipse approaches both sides of the square exponentially fast. All numerical results were checked in high-precision arithmetic with publicly available code.

Index Terms—Arc length, asymptotic expansion, Beta function, Gamma function, Mellin transform, superellipse.

I. INTRODUCTION

LENGTH is not continuous under uniform convergence. A staircase from $(0, 0)$ to $(1, 1)$ with ever finer steps converges uniformly to the diagonal, yet its length stays equal to 2 instead of approaching $\sqrt{2}$, because arc length is only lower semicontinuous [1].

The family $y = x^n$ on $[0, 1]$ is a smooth relative of the staircase. As $n \rightarrow \infty$ the graphs converge, in the Hausdorff sense, to the corner path

$$\Gamma : (0, 0) \rightarrow (1, 0) \rightarrow (1, 1),$$

and this time the lengths do converge to the length of the limit:

$$L(n) := \int_0^1 \sqrt{1 + n^2 x^{2n-2}} dx \rightarrow 2. \quad (1)$$

This limit is a standard exercise. In this paper we study the rate of convergence. We show that the error decays like $(\ln n)/n$, explain the origin of the logarithm, and give an exact expansion of the error in terms of the Gamma function. Fig. 1 shows the two families of curves studied in this paper.

A. Main Results

Throughout, $n > 1$ is real, $\ln$ denotes the natural logarithm, and

$$\varepsilon := \frac{1}{n-1}, \quad D(n) := 2 - L(n). \quad (2)$$

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TABLE I
NOTATION

Symbol	Meaning
$L(n)$	Arc length of $y = x^n$, $0 \leq x \leq 1$
ε	$1/(n-1)$
$D(n)$	Defect $2 - L(n)$
$g(s)$	Defect density $1 + s - \sqrt{1 + s^2}$
$k(s)$	$\sqrt{1 + s^2} - s = 1 - g(s)$
$M(\varepsilon)$	Mellin transform of k
$R(\varepsilon)$	Gamma-function factor (3)
$h_c(s)$	Defect density for $\cos \theta = c$ (11)
$L_\theta(n)$	Arc length of $\gamma_n = x u + x^n v$
$C(\theta)$	Constant term in Theorem 2
$Q(p), P(p)$	Quarter-arc length and perimeter of $ x ^p + y ^p = 1$
K, κ	First- and second-order superellipse constants

We call $D(n)$ the *defect*. Table I lists the notation used throughout. Our first result is elementary and has explicit constants.

Theorem A (Proposition 1): For every $n \geq 3$,

$$\begin{aligned} D(n) &\leq \frac{\ln n + 1}{n-1}, \\ D(n) &\geq \frac{\ln n - 1}{n-1} - \frac{\ln^2 n}{2(n-1)^2}. \end{aligned}$$

In particular, $D(n) \sim (\ln n)/n$ as $n \rightarrow \infty$.

The second result is exact. For $0 \leq \varepsilon < 1$ let

$$R(\varepsilon) := \frac{\Gamma(1 + \frac{\varepsilon}{2}) \Gamma(\frac{1-\varepsilon}{2})}{\sqrt{\pi} (1 + \varepsilon)}, \quad (3)$$

so that $R(0) = 1$.

Theorem B (Theorem 1 and Corollary 1): For every real $n > 2$,

$$\begin{aligned} L(n) &= 1 + \frac{R(\varepsilon)}{n^\varepsilon} \\ &\quad - \sum_{k=1}^{\infty} \binom{1/2}{k} \frac{1}{n^{2k-1} ((2k-1)(n-1) - 1)}. \end{aligned} \quad (4)$$

Consequently, as $n \rightarrow \infty$,

$$L(n) = 2 - \frac{\ln n + 1 - \ln 2}{n-1} + O\left(\frac{\ln^2 n}{n^2}\right), \quad (5)$$

$$L(n) = 1 + \left(\frac{2}{en}\right)^{\frac{1}{n-1}} + \frac{\pi^2}{12n^2} + O\left(\frac{\ln n}{n^3}\right). \quad (6)$$

The terms of the series in (4) decay like n^{-2k} , and four terms give $L(100)$ to about 22 digits. The error of the approximation (6) is below 5×10^{-6} at $n = 100$ and below 2×10^{-11} at $n = 10^4$ (Section V).

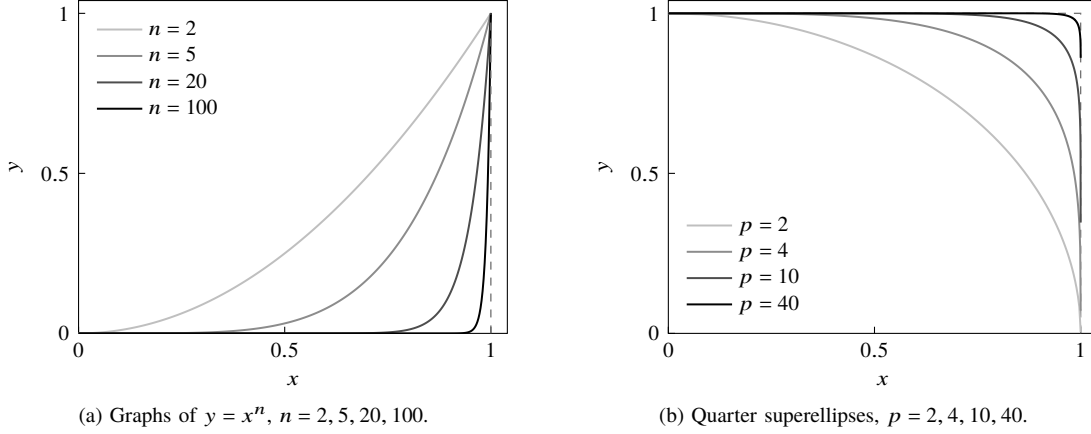


Fig. 1. Two families of curves that collapse onto a right-angled corner (dashed). (a) The length defect of $y = x^n$ is of order $(\ln n)/n$ (Theorems A and B). (b) The perimeter defect of the superellipse is of order $1/p$, with no logarithm (Theorem D).

TABLE II
SUMMARY OF THE MAIN RESULTS

Curve	Limit	Defect to leading order	Ref.
$y = x^n$	2	$(\ln n + 1 - \ln 2)/(n - 1)$	Thm. 1
$y = c x^n$	$1 + c$	$(\ln(cn) + 1 - \ln 2)/(n - 1)$	Cor. 2
$xu + x^n v$	2	$(2 \sin^2 \frac{\theta}{2} \ln n + C(\theta))/(n - 1)$	Thm. 2
$ x ^p + y ^p = 1$	8	$4K/p$	Thm. 3

Section VIII traces the logarithm to the slow approach of x^n to the vertical leg of the corner, and Section VI shows that it does not depend on the corner being a right angle. For unit vectors $u, v \in \mathbb{R}^2$ that make an angle $\theta \in [0, \pi)$, consider the curve

$$\gamma_n(x) = xu + x^n v, \quad 0 \leq x \leq 1, \quad (7)$$

which collapses onto the two-segment path $0 \rightarrow u \rightarrow u + v$ of length 2 as $n \rightarrow \infty$. The graph of x^n is the case $\theta = \pi/2$.

Theorem C (Theorem 2): Let $L_\theta(n)$ be the length of γ_n . Then

$$2 - L_\theta(n) = \frac{2 \sin^2 \frac{\theta}{2} \ln n + C(\theta)}{n - 1} + O\left(\frac{\ln^2 n}{n^2}\right),$$

where

$$C(\theta) = 2 \sin^2 \frac{\theta}{2} + 4 \cos^2 \frac{\theta}{2} \ln \cos \frac{\theta}{2}.$$

The logarithm does not appear for every curve that collapses onto a corner. The superellipse $|x|^p + |y|^p = 1$, popularized by Piet Hein [2], converges to the square $[-1, 1]^2$ as $p \rightarrow \infty$ at the faster rate $1/p$.

Theorem D (Theorem 3): The perimeter $P(p)$ of $|x|^p + |y|^p = 1$ satisfies, for every $p \geq 2$,

$$\left| P(p) - 8 + \frac{4K}{p} \right| \leq \frac{12}{p^2},$$

where $K = 2\sqrt{2} \ln(1 + \sqrt{2}) - 2 \ln 2 = 1.10660 \dots$

Table II collects these results, together with the scaled curves $y = c x^n$ treated in Corollary 2.

B. Related Work

The limit $L(n) \rightarrow 2$ is standard. The indefinite arc-length integral of x^n is a Gauss hypergeometric function; see the discussion in [3], which gives in particular

$$L(n) = {}_2F_1\left(-\frac{1}{2}, \frac{\varepsilon}{2}; 1 + \frac{\varepsilon}{2}; -n^2\right). \quad (8)$$

In principle, (4) can be obtained from (8) through the classical connection formula that transforms a ${}_2F_1$ at argument z into functions of $1/z$ [4, Eq. 15.8.2], [5, Ch. 2]; see Remark 3. Our derivation does not use hypergeometric functions. It needs one substitution and one Beta integral, and it leads directly to the asymptotic statements (5) and (6), which we have not found in the literature.

Two recent preprints [6], [7] give hypergeometric series for the arc length of supercircles and for the perimeter of Lamé superellipses, and they verify the limiting value 8 as $p \rightarrow \infty$. They do not give the rate $4K/p$ of Theorem D, and we have not found it elsewhere.

The integrals behind our results have the form $\int_0^n h(s) s^{\varepsilon-1} ds$ with a small parameter ε , which is the natural setting of Mellin-transform asymptotics [8], [9]; see also [10] for the classical methods. We need only one Mellin transform, which we evaluate directly. Because these questions are elementary, some of our results may already appear in the problem sections of journals, and we would be grateful for references.

C. Contributions and Organization

The contributions of this paper are as follows.

- A reduction of each problem to a single one-dimensional integral, by comparing Euclidean with taxicab length (Section II).
- Explicit two-sided bounds for the defect $D(n)$ (Section III).
- An exact, rapidly convergent Gamma-function series for $L(n)$, derived from one Beta integral (Section IV), and its asymptotic expansion to any order (Section V and Appendix A).

- The extension to corners of arbitrary angle, with the constant term in closed form (Section VI).
- A contrasting example without a logarithm, the superellipse, with an explicit constant and an explicit error bound (Section VII).
- A geometric explanation of when the logarithm appears (Section VIII), a list of open problems (Section IX), and a reproducible high-precision verification of every numerical claim (Appendix B).

II. THE TAXICAB DEFECT

We compare the Euclidean length of a curve with its *taxicab* length. Let $\gamma(t) = (x(t), y(t))$ be a C^1 curve from $(0, 0)$ to $(1, 1)$ whose two coordinates are monotone. Its taxicab length is $\int (|x'| + |y'|) dt = 1 + 1 = 2$, so

$$2 - \text{length}(\gamma) = \int \left(|x'| + |y'| - \sqrt{x'^2 + y'^2} \right) dt \geq 0. \quad (9)$$

The integrand vanishes where the curve is horizontal or vertical and is largest where the curve moves diagonally, so the defect measures how far the curve travels in diagonal directions. The same comparison works for any pair of directions.

Lemma 1 (Defect integral): Let $u, v \in \mathbb{R}^2$ be unit vectors with $u \cdot v = c = \cos \theta$, where $\theta \in [0, \pi)$, and let γ_n be as in (7). Then

$$2 - \text{length}(\gamma_n) = \varepsilon n^{-\varepsilon} \int_0^n h_c(s) s^{\varepsilon-1} ds, \quad (10)$$

where, for $s \geq 0$,

$$\begin{aligned} h_c(s) &:= 1 + s - \sqrt{1 + 2cs + s^2} \\ &= \frac{2(1-c)s}{1 + s + \sqrt{1 + 2cs + s^2}}. \end{aligned} \quad (11)$$

Proof: Write $y = x^n$. Then $\gamma'_n = u + y'v$ and $|\gamma'_n| = \sqrt{1 + 2cy' + y'^2}$. Since $\int_0^1 y' dx = 1$,

$$\begin{aligned} 2 - \text{length}(\gamma_n) &= \int_0^1 (1 + y' - |u + y'v|) dx \\ &= \int_0^1 h_c(nx^{n-1}) dx. \end{aligned}$$

Now substitute $s = nx^{n-1}$, an increasing bijection from $(0, 1]$ onto $(0, n]$, with $x = (s/n)^\varepsilon$ and $dx = \varepsilon (s/n)^\varepsilon ds/s$. This gives (10). The second expression in (11) follows by multiplying and dividing by $1 + s + \sqrt{1 + 2cs + s^2}$. ■

The graph of x^n is the case $u = (1, 0)$, $v = (0, 1)$, $c = 0$. We write

$$g(s) := h_0(s) = 1 + s - \sqrt{1 + s^2}, \quad (12)$$

so that

$$D(n) = \varepsilon n^{-\varepsilon} \int_0^n g(s) s^{\varepsilon-1} ds. \quad (13)$$

The function g is the length lost per unit of horizontal distance along a stretch of slope s (Fig. 2). Two properties of g are used below. For small s , $g(s) \approx s$, so gentle slopes lose little length. As $s \rightarrow \infty$, $g(s) \rightarrow 1$, so steep slopes lose almost the whole horizontal step. The second property makes $\int_1^n g(s) ds/s$ grow like $\ln n$.

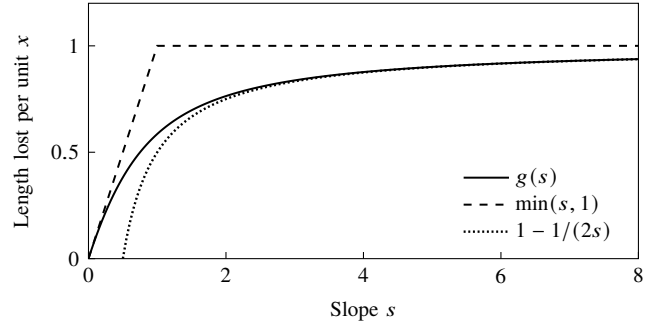


Fig. 2. The defect density $g(s) = 1 + s - \sqrt{1 + s^2}$, which is the length lost per unit of horizontal distance along a stretch of slope s , with the bounds (14) and (15). The loss is small for gentle slopes and approaches the full horizontal step for steep ones.

III. ELEMENTARY BOUNDS

Proposition 1: For every real $n \geq 3$,

$$\begin{aligned} D(n) &\leq \frac{\ln n + 1}{n - 1}, \\ D(n) &\geq \frac{\ln n - 1}{n - 1} - \frac{\ln^2 n}{2(n - 1)^2}. \end{aligned}$$

Proof: We use two bounds on g . Since $1 + s + \sqrt{1 + s^2} \geq 1 + s + \max(1, s) \geq 2 \max(1, s)$, the second form of g in (11) gives

$$0 \leq g(s) \leq \min(s, 1). \quad (14)$$

Also $g(s) = 1 - (\sqrt{1 + s^2} - s) = 1 - 1/(s + \sqrt{1 + s^2})$, hence

$$g(s) \geq 1 - \frac{1}{2s} \quad (s > 0). \quad (15)$$

Upper bound. By (14),

$$\begin{aligned} \int_0^n g(s) s^{\varepsilon-1} ds &\leq \int_0^1 s^{\varepsilon} ds + \int_1^n s^{\varepsilon-1} ds \\ &= \frac{1}{1 + \varepsilon} + \frac{n^{\varepsilon} - 1}{\varepsilon}. \end{aligned}$$

Multiplying by $\varepsilon n^{-\varepsilon}$ gives $D(n) \leq \varepsilon + (1 - n^{-\varepsilon}) \leq \varepsilon + \varepsilon \ln n$, because $1 - e^{-x} \leq x$.

Lower bound. Drop the nonnegative integral over $[0, 1]$ and apply (15) on $[1, n]$:

$$\begin{aligned} D(n) &\geq \varepsilon n^{-\varepsilon} \int_1^n \left(s^{\varepsilon-1} - \frac{1}{2} s^{\varepsilon-2} \right) ds \\ &= (1 - n^{-\varepsilon}) - \frac{\varepsilon n^{-\varepsilon}}{2} \cdot \frac{1 - n^{-\varepsilon-1}}{1 - \varepsilon} \\ &\geq (1 - n^{-\varepsilon}) - \frac{\varepsilon}{2(1 - \varepsilon)}. \end{aligned}$$

For $n \geq 3$ we have $\varepsilon \leq \frac{1}{2}$, so $\varepsilon/(2(1 - \varepsilon)) \leq \varepsilon$. Finally, $1 - e^{-x} \geq x - x^2/2$ for $x \geq 0$; with $x = \varepsilon \ln n$ this gives the claim. ■

Proposition 1 already shows that $n(2 - L(n))/\ln n \rightarrow 1$. Its bounds leave the constant next to $\ln n$ undetermined within $[-1, 1]$. Section V shows that this constant equals $1 - \ln 2 \approx 0.307$.

IV. AN EXACT FORMULA

The function $k(s) := \sqrt{1+s^2} - s = 1 - g(s)$ is positive and decreasing, with $k(0) = 1$ and $k(s) \leq 1/(2s)$. Its Mellin transform

$$M(\varepsilon) := \int_0^\infty k(s) s^{\varepsilon-1} ds \quad (16)$$

therefore converges for $0 < \varepsilon < 1$. Mellin transforms are the standard tool for integrals such as (13) [8], [9]; here we need only the evaluation in Lemma 2.

Lemma 2: For $0 < \varepsilon < 1$, we have $\varepsilon M(\varepsilon) = R(\varepsilon)$, with R as in (3).

Proof: Put $w = k(s) = \sqrt{1+s^2} - s$. As s runs over $(0, \infty)$, w decreases from 1 to 0. Solving gives $s = (1-w^2)/(2w)$ and $ds = -\frac{1+w^2}{2w^2} dw$. Thus

$$\begin{aligned} M(\varepsilon) &= \int_0^1 w \left(\frac{1-w^2}{2w} \right)^{\varepsilon-1} \frac{1+w^2}{2w^2} dw \\ &= 2^{-\varepsilon} \int_0^1 (1-w^2)^{\varepsilon-1} w^{-\varepsilon} (1+w^2) dw. \end{aligned}$$

Substituting $w = \sqrt{t}$ gives Euler Beta integrals $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$ [4, Eq. 5.12.1]:

$$\begin{aligned} M(\varepsilon) &= 2^{-\varepsilon-1} \int_0^1 (1-t)^{\varepsilon-1} (t^{-\frac{1+\varepsilon}{2}} + t^{\frac{1-\varepsilon}{2}}) dt \\ &= 2^{-\varepsilon-1} \left[B\left(\frac{1-\varepsilon}{2}, \varepsilon\right) + B\left(\frac{3-\varepsilon}{2}, \varepsilon\right) \right]. \end{aligned}$$

Write $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ and use $\Gamma(\frac{3\pm\varepsilon}{2}) = \frac{1\pm\varepsilon}{2}\Gamma(\frac{1\pm\varepsilon}{2})$. The bracket becomes

$$\frac{\Gamma(\varepsilon)\Gamma(\frac{1-\varepsilon}{2})}{\Gamma(\frac{1+\varepsilon}{2})} \left(1 + \frac{1-\varepsilon}{1+\varepsilon}\right) = \frac{\Gamma(\varepsilon)\Gamma(\frac{1-\varepsilon}{2})}{\Gamma(\frac{3+\varepsilon}{2})}.$$

Hence $\varepsilon M(\varepsilon) = \Gamma(1+\varepsilon)\Gamma(\frac{1-\varepsilon}{2})/(2^{1+\varepsilon}\Gamma(\frac{3+\varepsilon}{2}))$. Legendre's duplication formula $\Gamma(1+\varepsilon) = \pi^{-1/2}2^\varepsilon\Gamma(\frac{1+\varepsilon}{2})\Gamma(1+\frac{\varepsilon}{2})$ [4, Eq. 5.5.5], [5, Ch. 1], together with $\Gamma(\frac{3+\varepsilon}{2}) = \frac{1+\varepsilon}{2}\Gamma(\frac{1+\varepsilon}{2})$, turns this into $R(\varepsilon)$. ■

Theorem 1 (Exact formula): For every real $n > 2$, with $\varepsilon = 1/(n-1)$,

$$\begin{aligned} L(n) &= 1 + \frac{R(\varepsilon)}{n^\varepsilon} \\ &\quad - \sum_{k=1}^\infty \binom{1/2}{k} \frac{1}{n^{2k-1}((2k-1)(n-1)-1)}, \quad (17) \end{aligned}$$

where $\binom{1/2}{k} = \frac{1}{k!} \prod_{j=0}^{k-1} (\frac{1}{2} - j)$. The coefficients are $\frac{1}{2}, -\frac{1}{8}, \frac{1}{16}, -\frac{5}{128}, \dots$, and the series converges absolutely.

Proof: Since $n > 2$, we have $0 < \varepsilon < 1$. By (13) and $g = 1 - k$,

$$\begin{aligned} \int_0^n g(s) s^{\varepsilon-1} ds &= \frac{n^\varepsilon}{\varepsilon} - \int_0^\infty k(s) s^{\varepsilon-1} ds \\ &\quad + \int_n^\infty k(s) s^{\varepsilon-1} ds. \end{aligned}$$

Multiplying by $\varepsilon n^{-\varepsilon}$ and applying Lemma 2 gives

$$D(n) = 1 - \frac{R(\varepsilon)}{n^\varepsilon} + \varepsilon n^{-\varepsilon} \int_n^\infty k(s) s^{\varepsilon-1} ds. \quad (18)$$

For $s > 1$, the binomial series gives

$$k(s) = s(\sqrt{1+s^{-2}} - 1) = \sum_{k \geq 1} \binom{1/2}{k} s^{1-2k}.$$

Because $\sum_k |\binom{1/2}{k}| n^{-2k} < \infty$ for $n > 1$, we may integrate term by term on $[n, \infty)$ by Tonelli's theorem. Each term contributes $\int_n^\infty s^{\varepsilon-2k} ds = n^{\varepsilon+1-2k}/(2k-1-\varepsilon)$. Hence

$$\begin{aligned} \varepsilon n^{-\varepsilon} \int_n^\infty k(s) s^{\varepsilon-1} ds &= \sum_{k \geq 1} \binom{1/2}{k} \frac{\varepsilon n^{1-2k}}{2k-1-\varepsilon} \\ &= \sum_{k \geq 1} \binom{1/2}{k} \frac{1}{n^{2k-1}((2k-1)(n-1)-1)}. \end{aligned}$$

Finally, $L(n) = 2 - D(n)$. ■

Example 1 ($n = 3$): Here $\varepsilon = \frac{1}{2}$ and $R(\frac{1}{2}) = \Gamma(\frac{5}{4})\Gamma(\frac{1}{4})/(\frac{3}{2}\sqrt{\pi}) = \Gamma(\frac{1}{4})^2/(6\sqrt{\pi})$. Theorem 1 gives the length of $y = x^3$ on $[0, 1]$ as

$$L(3) = 1 + \frac{\Gamma(\frac{1}{4})^2}{6\sqrt{3\pi}} - \sum_{k \geq 1} \binom{1/2}{k} \frac{1}{3^{2k-1}(4k-3)} = 1.547865\dots,$$

in agreement with direct quadrature of $\int_0^1 \sqrt{1+9x^4} dx$ to 40 digits. The Gamma-function term equals $0.713634\dots$, and the series converges geometrically with ratio $1/9$.

Remark 1 (The case $n = 2$): At $n = 2$ we have $\varepsilon = 1$. Both $R(\varepsilon)$, through $\Gamma(\frac{1-\varepsilon}{2})$, and the $k = 1$ term of the series have poles, and the two poles cancel. Since L is continuous in n , the limit $n \rightarrow 2^+$ of (17) is the length of the parabola, $L(2) = \frac{\sqrt{5}}{2} + \frac{1}{4} \ln(2 + \sqrt{5})$.

Remark 2 (Scaled curves): The same proof applies to $y = c x^n$ on $[0, 1]$ for any $c > 0$ with $cn > 1$; the only change is that the substitution now ends at $s = cn$. Writing $\ell_c(n)$ for its length,

$$\begin{aligned} \ell_c(n) &= c + \frac{R(\varepsilon)}{(cn)^\varepsilon} \\ &\quad - \sum_{k \geq 1} \binom{1/2}{k} \frac{1}{(cn)^{2k-1}((2k-1)(n-1)-1)}. \quad (19) \end{aligned}$$

Scaling shows that the length of $y = x^n$ on $[0, a]$ equals $a \ell_{a^{n-1}}(n)$, so this also covers every interval $[0, a]$.

Remark 3 (Relation to the hypergeometric form): The large-argument connection formula [4, Eq. 15.8.2] expresses (8) as a combination of two ${}_2F_1$'s at argument $-1/n^2$. One can check that the Gamma factor multiplying $n^{-\varepsilon}$ there, $\Gamma(1+\frac{\varepsilon}{2})\Gamma(-\frac{1+\varepsilon}{2})/\Gamma(-\frac{1}{2})$, equals $R(\varepsilon)$. The derivation given here is elementary, and the form (17) leads directly to the asymptotic results of the next section.

V. ASYMPTOTIC EXPANSION

We need the Taylor expansion of $\ln R$ at 0. It uses the classical values $\psi(1) = -\gamma$, $\psi(\frac{1}{2}) = -\gamma - 2 \ln 2$, $\psi'(1) = \pi^2/6$, $\psi'(\frac{1}{2}) = \pi^2/2$, $\psi''(1) = -2\zeta(3)$, and $\psi''(\frac{1}{2}) = -14\zeta(3)$, where $\psi = \Gamma'/\Gamma$ and γ is Euler's constant [4, Secs. 5.4 and 5.15].

Lemma 3: As $\varepsilon \rightarrow 0$,

$$\ln R(\varepsilon) = a_1 \varepsilon + a_2 \varepsilon^2 + a_3 \varepsilon^3 + O(\varepsilon^4),$$

where $a_1 = \ln 2 - 1$, $a_2 = \frac{\pi^2}{12} + \frac{1}{2}$, and $a_3 = \frac{\zeta(3)}{4} - \frac{1}{3}$.

Proof: We have $\ln R(\varepsilon) = \ln \Gamma(1 + \frac{\varepsilon}{2}) + \ln \Gamma(\frac{1-\varepsilon}{2}) - \frac{1}{2} \ln \pi - \ln(1 + \varepsilon)$, and R is analytic near 0. The j th Taylor coefficient of $\ln \Gamma(\alpha + \beta\varepsilon)$ is $\beta^j \psi^{(j-1)}(\alpha)/j!$. Hence

$$\begin{aligned} a_1 &= \frac{1}{2} \psi(1) - \frac{1}{2} \psi\left(\frac{1}{2}\right) - 1 = \ln 2 - 1, \\ a_2 &= \frac{1}{8} \psi'(1) + \frac{1}{8} \psi'\left(\frac{1}{2}\right) + \frac{1}{2} = \frac{\pi^2}{48} + \frac{\pi^2}{16} + \frac{1}{2}, \\ a_3 &= \frac{1}{48} \psi''(1) - \frac{1}{48} \psi''\left(\frac{1}{2}\right) - \frac{1}{3} = \frac{12\zeta(3)}{48} - \frac{1}{3}. \end{aligned}$$

Euler's constant cancels in a_1 . ■

Corollary 1 (Asymptotics of $L(n)$): As $n \rightarrow \infty$,

$$\begin{aligned} \text{(a)} \quad L(n) &= 2 - \frac{\ln n + 1 - \ln 2}{n-1} + O\left(\frac{\ln^2 n}{n^2}\right), \\ \text{(b)} \quad L(n) &= 1 + \left(\frac{2}{en}\right)^{\frac{1}{n-1}} + \frac{\pi^2}{12n^2} + O\left(\frac{\ln n}{n^3}\right). \end{aligned}$$

Proof: Put $\lambda = \ln n + 1 - \ln 2 = \ln n - a_1$. By Lemma 3,

$$\begin{aligned} \frac{R(\varepsilon)}{n^\varepsilon} &= \exp(-\varepsilon\lambda + a_2\varepsilon^2 + O(\varepsilon^3)) \\ &= e^{-\varepsilon\lambda} (1 + a_2\varepsilon^2 + O(\varepsilon^3)). \end{aligned}$$

The series in Theorem 1 equals $\frac{1}{2n(n-2)} + O(n^{-4})$.

For (a), note that $e^{-\varepsilon\lambda} = 1 - \varepsilon\lambda + O(\varepsilon^2 \ln^2 n)$, and that every other term is $O(n^{-2})$.

For (b), note that $e^{-\varepsilon\lambda} = 1 + O(\varepsilon \ln n)$, so $e^{-\varepsilon\lambda} a_2 \varepsilon^2 = a_2 \varepsilon^2 + O(\ln n/n^3)$. Hence

$$L(n) = 1 + e^{-\varepsilon\lambda} + a_2 \varepsilon^2 - \frac{1}{2n(n-2)} + O\left(\frac{\ln n}{n^3}\right).$$

Now $a_2 \varepsilon^2 = a_2/n^2 + O(n^{-3})$, $\frac{1}{2n(n-2)} = \frac{1}{2n^2} + O(n^{-3})$, and $a_2 - \frac{1}{2} = \frac{\pi^2}{12}$. Finally, $e^{-\varepsilon\lambda} = \exp\left(-\frac{\ln n + 1 - \ln 2}{n-1}\right) = \left(\frac{2}{en}\right)^{1/(n-1)}$. ■

The same argument, carried one step further, yields the next two terms (Proposition 2 in Appendix A). All further terms can be computed in the same way; they are polynomials in $\ln n$ divided by powers of n .

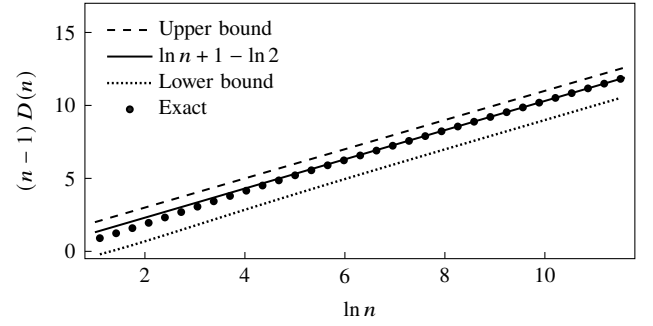
Corollary 2 (Scaled curves): Fix $c > 0$, and let $\ell_c(n)$ be the length of $y = cx^n$ on $[0, 1]$, which tends to $1 + c$. As $n \rightarrow \infty$,

$$1 + c - \ell_c(n) = \frac{\ln(cn) + 1 - \ln 2}{n-1} + O\left(\frac{\ln^2 n}{n^2}\right).$$

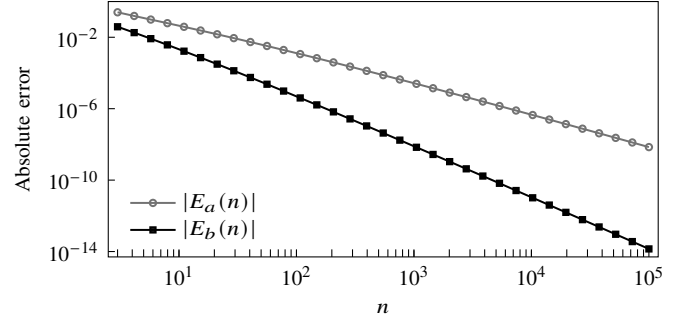
Proof: By (19), $1 + c - \ell_c(n) = 1 - (cn)^{-\varepsilon} R(\varepsilon) + S_c(n)$, where the series $S_c(n)$ equals $\frac{1}{2cn(n-2)} + O(n^{-4}) = O(n^{-2})$ for fixed c . By Lemma 3, $(cn)^{-\varepsilon} R(\varepsilon) = \exp(-\varepsilon(\ln(cn) + 1 - \ln 2) + O(\varepsilon^2))$, which equals $1 - \varepsilon(\ln(cn) + 1 - \ln 2) + O(\varepsilon^2 \ln^2 n)$. ■

Stretching the vertical leg of the corner by a factor c therefore changes the defect only through $\ln c$, by $(\ln c)/(n-1)$ to first order.

Fig. 3 and Table III compare Corollary 1 with exact values. Write $E_a(n)$ and $E_b(n)$ for the errors of the approximations in parts (a) and (b), that is, $L(n)$ minus the right-hand side without its O -term. Fig. 3(a) plots the rescaled defect $(n-1)D(n)$ against $\ln n$. The exact values lie on the line $\ln n + 1 - \ln 2$, between the bounds of Proposition 1. Fig. 3(b) compares $|E_a(n)|$ and $|E_b(n)|$.



(a) Rescaled defect $(n-1)D(n)$ against $\ln n$.



(b) Absolute errors $|E_a(n)|$ and $|E_b(n)|$ of Corollary 1.

Fig. 3. Accuracy of Corollary 1. (a) The rescaled defect, computed from Theorem 1, follows the line $\ln n + 1 - \ln 2$ and stays between the bounds of Proposition 1. (b) The error of approximation (a) decays like $\ln^2 n/n^2$; the error of approximation (b) decays like $\ln n/n^3$.

TABLE III
ARC LENGTH OF $y = x^n$ AND THE ERRORS OF COROLLARY 1

n	$L(n)$	$E_a(n)$	$E_b(n)$
3	1.547865654684	2.5×10^{-1}	-3.9×10^{-2}
5	1.640559757853	1.2×10^{-1}	-1.2×10^{-2}
10	1.754409376495	4.4×10^{-2}	-2.1×10^{-3}
20	1.842143162816	1.6×10^{-2}	-3.6×10^{-4}
50	1.917799927761	3.9×10^{-3}	-3.2×10^{-5}
100	1.951671753851	1.3×10^{-3}	-4.9×10^{-6}
1000	1.992804999370	2.7×10^{-5}	-7.9×10^{-9}
10000	1.999048646545	4.6×10^{-7}	-1.1×10^{-11}

VI. CORNERS OF ARBITRARY ANGLE

The leading term in Corollary 1(a) depends only on two properties of g : it is roughly linear near 0, and it has a limit at ∞ . Lemma 4 treats any function with these two properties.

Lemma 4: Let $h : [0, \infty) \rightarrow \mathbb{R}$ be continuous. Suppose there are constants $A, B \geq 0$ and h_∞ such that $|h(s)| \leq As$ for $0 \leq s \leq 1$ and $|h(s) - h_\infty| \leq B/s$ for $s \geq 1$. Put

$$C_h := \int_0^1 \frac{h(s)}{s} ds + \int_1^\infty \frac{h(s) - h_\infty}{s} ds.$$

Then for $n \geq 3$ and $\varepsilon = 1/(n-1)$,

$$\begin{aligned} \varepsilon n^{-\varepsilon} \int_0^n h(s) s^{\varepsilon-1} ds \\ = \varepsilon(h_\infty \ln n + C_h) + O(\varepsilon^2 \ln^2 n), \end{aligned}$$

where the implied constant depends only on A, B , and h_∞ .

Proof: Split $J := \int_0^n h(s)s^{\varepsilon-1} ds$ as

$$J = \int_0^1 h(s)s^{\varepsilon-1} ds + h_\infty \int_1^n s^{\varepsilon-1} ds + \int_1^n (h(s) - h_\infty)s^{\varepsilon-1} ds.$$

We compare each piece with its value at $\varepsilon = 0$. For $0 < s \leq 1$ we have $|s^\varepsilon - 1| \leq \varepsilon |\ln s|$, so the first piece differs from $\int_0^1 h(s) ds/s$ by at most $A\varepsilon \int_0^1 |\ln s| ds = A\varepsilon$. For $1 \leq s \leq n$ we have $|s^\varepsilon - 1| \leq \varepsilon \ln s \cdot n^\varepsilon \leq e\varepsilon \ln s$, because $n^\varepsilon = e^{\ln n/(n-1)} \leq e$. So the third piece differs from $\int_1^\infty (h - h_\infty) ds/s$ by at most $eB\varepsilon \int_1^\infty s^{-2} \ln s ds + \int_n^\infty Bs^{-2} ds \leq eB\varepsilon + B/n$. Therefore

$$J = h_\infty \frac{n^\varepsilon - 1}{\varepsilon} + C_h + E, \quad |E| \leq (A + eB + B)\varepsilon.$$

Multiplying by $\varepsilon n^{-\varepsilon}$ gives $h_\infty(1 - n^{-\varepsilon}) + \varepsilon n^{-\varepsilon}(C_h + E)$. Put $x = \varepsilon \ln n \in (0, 1]$. Then $1 - e^{-x} = x + O(x^2)$ and $e^{-x} = 1 + O(x)$, and since $\ln n \geq 1$, every error term is $O(\varepsilon^2 \ln^2 n)$. ■

Theorem 2 (Corners of arbitrary angle): Let u, v be unit vectors at angle $\theta \in [0, \pi)$, and let $L_\theta(n)$ be the length of $\gamma_n(x) = xu + x^n v$, $0 \leq x \leq 1$. Then

$$2 - L_\theta(n) = \frac{2 \sin^2 \frac{\theta}{2} \ln n + C(\theta)}{n - 1} + O\left(\frac{\ln^2 n}{n^2}\right),$$

where

$$C(\theta) = 2 \sin^2 \frac{\theta}{2} + 4 \cos^2 \frac{\theta}{2} \ln \cos \frac{\theta}{2}. \quad (20)$$

For $\theta = \pi/2$ this gives $C = 1 - \ln 2$, in agreement with Corollary 1(a). For $\theta = 0$ both terms vanish, since γ_n is then a straight segment. The coefficient $2 \sin^2(\theta/2) = 1 - \cos \theta$ increases with the turning angle.

Proof: By Lemma 1, we must apply Lemma 4 to $h = h_c$ with $c = \cos \theta \in (-1, 1]$. Write $r(s) = \sqrt{1 + 2cs + s^2}$. The second form of (11) gives $0 \leq h_c(s) \leq (1 - c)s$, so $A = 1 - c$ works. A short computation gives

$$h_c(s) - (1 - c) = \frac{(1 - c)(s - 1 - r(s))}{1 + s + r(s)}.$$

Since $|s + c| \leq r(s) \leq s + 1$, we have $|s - 1 - r(s)| \leq 2$. Therefore $h_\infty = 1 - c$ and $B = 2(1 - c)$ work.

It remains to compute C_{h_c} . Differentiation confirms that

$$G(s) = r(s) + c \ln(s + c + r(s)) - \ln \frac{1 + cs + r(s)}{s}$$

satisfies $G'(s) = r(s)/s$. (Since $r(s) \geq |s + c|$ and $r(s) \geq |1 + cs|$, with $c > -1$, both logarithms are defined for $s > 0$.) Hence $\int_\delta^N h_c(s) ds/s = [\ln s + s - G(s)]_\delta^N$, and

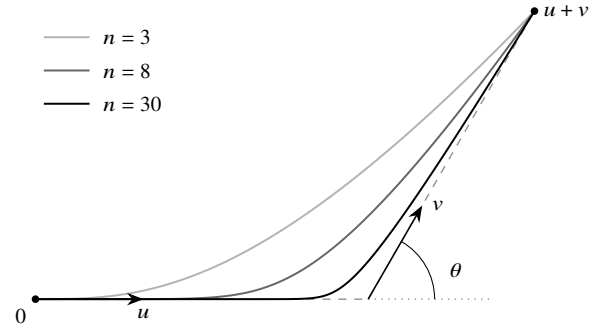
$$C_{h_c} = \lim_{\substack{\delta \rightarrow 0 \\ N \rightarrow \infty}} \left(\int_\delta^N \frac{h_c(s)}{s} ds - (1 - c) \ln N \right).$$

As $N \rightarrow \infty$, we have $r(N) = N + c + O(1/N)$, $N + c + r(N) \sim 2N$, and $(1 + cN + r(N))/N \rightarrow 1 + c$. Thus

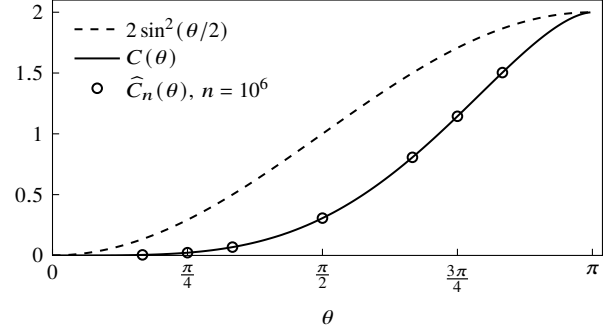
$$\ln N + N - G(N) = (1 - c) \ln N - c - c \ln 2 + \ln(1 + c) + o(1).$$

As $\delta \rightarrow 0$, we have $r(\delta) \rightarrow 1$ and $(1 + c\delta + r(\delta))/\delta \sim 2/\delta$. Thus

$$\ln \delta + \delta - G(\delta) = -1 - c \ln(1 + c) + \ln 2 + o(1).$$



(a) The curves γ_n for $\theta = \pi/3$ and $n = 3, 8, 30$.



(b) The coefficient $2 \sin^2(\theta/2)$ and the constant $C(\theta)$.

Fig. 4. Corners of arbitrary angle. (a) As n grows, γ_n collapses onto the path $0 \rightarrow u \rightarrow u + v$ (dashed), which turns by the angle θ . (b) The coefficient of $\ln n/(n-1)$ and the constant term in Theorem 2; the circles are the empirical constants of Table IV.

TABLE IV
NUMERICAL CHECK OF THEOREM 2 AT $n = 10^6$

θ	$2 \sin^2(\theta/2)$	$C(\theta)$	$\hat{C}_n(\theta)$
$\pi/6$	0.133975	0.0045910	0.0045781
$\pi/4$	0.292893	0.0225777	0.0225492
$\pi/3$	0.500000	0.0684769	0.0684278
$\pi/2$	1.000000	0.3068528	0.3067523
$2\pi/3$	1.500000	0.8068528	0.8066971
$3\pi/4$	1.707107	1.1444313	1.1442509
$5\pi/6$	1.866025	1.5038583	1.5036576

Subtracting gives

$$C_{h_c} = (1 - c) + (1 + c) \ln \frac{1 + c}{2}.$$

Finally, $1 - c = 2 \sin^2 \frac{\theta}{2}$ and $\frac{1+c}{2} = \cos^2 \frac{\theta}{2}$. ■

Fig. 4(a) shows the curves γ_n for $\theta = \pi/3$. To test Theorem 2, define the empirical constant

$$\hat{C}_n(\theta) := (n - 1)(2 - L_\theta(n)) - 2 \sin^2 \frac{\theta}{2} \ln n,$$

which by the theorem equals $C(\theta) + O(\ln^2 n/n)$. Table IV and Fig. 4(b) compare $\hat{C}_n(\theta)$ at $n = 10^6$, computed by direct quadrature, with the closed form (20). The differences are about 10^{-4} , which is consistent with the error term.

VII. A CORNER WITHOUT A LOGARITHM: THE SUPERELLIPSE

Let $Q(p)$ be the length of the arc $x^p + y^p = 1$, $x, y \geq 0$, and let $P(p) = 4Q(p)$ be the perimeter of $|x|^p + |y|^p = 1$. At $p = 2$, $P = 2\pi$.

Theorem 3 (Superellipse perimeter): Let $K := 2\sqrt{2} \ln(1 + \sqrt{2}) - 2 \ln 2 = 1.1066065994 \dots$. For every $p \geq 2$,

$$\left| Q(p) - 2 + \frac{K}{p} \right| \leq \frac{3}{p^2},$$

$$\left| P(p) - 8 + \frac{4K}{p} \right| \leq \frac{12}{p^2}.$$

Proof: Parametrize the arc by $t \in (0, 1)$ via $x = t^{1/p}$ and $y = (1-t)^{1/p}$. Put $\varepsilon = 1/p \leq \frac{1}{2}$, $a = t^{\varepsilon-1}$, and $b = (1-t)^{\varepsilon-1}$. Then $|x'| = a/p$ and $|y'| = b/p$, and by (9),

$$2 - Q(p) = \frac{1}{p} F(\varepsilon), \quad F(\varepsilon) := \int_0^1 \Phi_\varepsilon(t) dt,$$

where $\Phi_\varepsilon := a + b - \sqrt{a^2 + b^2}$.

Step 1: $F(0) = K$. At $\varepsilon = 0$, $a = 1/t$ and $b = 1/(1-t)$. Put $q = \sqrt{t^2 + (1-t)^2}$. Then $\Phi_0 = (1-q)/(t(1-t))$, and $1-q = (1-q^2)/(1+q) = 2t(1-t)/(1+q)$, so the integrand simplifies to $\Phi_0(t) = 2/(1+q)$. Substitute $w = 2t-1$, so that $q^2 = (1+w^2)/2$, and then $w = \sinh \phi$. With $A = \operatorname{arsinh} 1 = \ln(1 + \sqrt{2})$,

$$F(0) = 2 \int_0^1 \frac{dw}{1 + \sqrt{(1+w^2)/2}} = 2\sqrt{2} \int_0^A \frac{\cosh \phi}{\sqrt{2} + \cosh \phi} d\phi$$

$$= 2\sqrt{2} A - 4 \int_0^A \frac{d\phi}{\sqrt{2} + \cosh \phi}.$$

With $\tau = \tanh(\phi/2)$ and $\alpha = \sqrt{2} + 1$, and using $\tanh(A/2) = \sinh A/(1 + \cosh A) = \sqrt{2} - 1$, the last integral becomes

$$\int_0^{\sqrt{2}-1} \frac{2 d\tau}{(\sqrt{2}+1) - (\sqrt{2}-1)\tau^2} = \ln \frac{\alpha + \tau}{\alpha - \tau} \Big|_{\tau=\sqrt{2}-1} = \frac{1}{2} \ln 2.$$

Hence $F(0) = K$.

Step 2: $|F(\varepsilon) - K| \leq 3\varepsilon$. Since $\Phi_\varepsilon(t) = \Phi_\varepsilon(1-t)$, it suffices to consider $0 < t \leq \frac{1}{2}$. Write $r = \sqrt{a^2 + b^2}$. Then $\partial \Phi / \partial a = 1 - a/r \in [0, 1]$, and likewise for b . Also $\partial_\varepsilon a = a \ln t$ and $\partial_\varepsilon b = b \ln(1-t)$. For $t \leq \frac{1}{2}$ and $\varepsilon \leq \frac{1}{2}$ we have $1 \leq b \leq 2$ and $a \geq t^{-1/2}$. Moreover $(1 - a/r)a = ab^2/(r(r+a)) \leq b^2/(2a)$. Hence

$$|\partial_\varepsilon \Phi_\varepsilon(t)| \leq \frac{b^2}{2a} |\ln t| + b |\ln(1-t)|$$

$$\leq 2\sqrt{t} |\ln t| + 2 \ln 2 \leq \frac{4}{e} + 2 \ln 2 < 3.$$

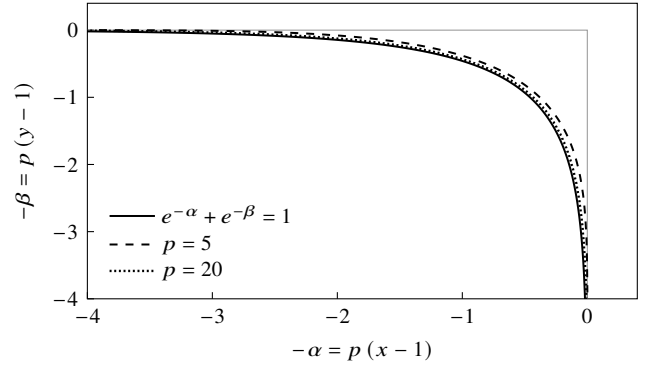
By the mean value theorem, $|\Phi_\varepsilon(t) - \Phi_0(t)| \leq 3\varepsilon$ for every t . Integrating over $[0, 1]$ gives $|F(\varepsilon) - K| \leq 3\varepsilon$. The result follows from $Q(p) = 2 - F(1/p)/p$. ■

Numerical evaluation of $F'(0) = \int_0^1 \partial_\varepsilon \Phi_\varepsilon(t)|_{\varepsilon=0} dt$ gives $F'(0) = -0.5052381924 \dots$. This indicates that

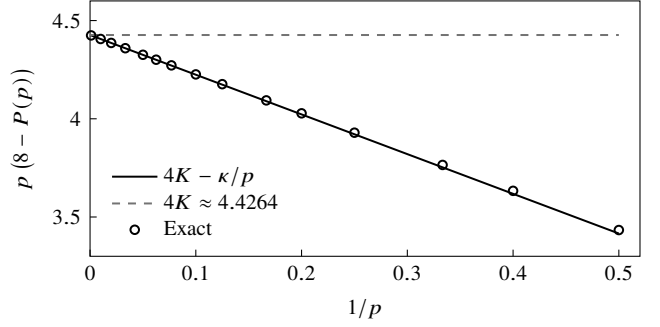
$$P(p) = 8 - \frac{4K}{p} + \frac{\kappa}{p^2} + o(p^{-2}), \quad (21)$$

with $\kappa = -4F'(0) = 2.0209527696 \dots$, well within the bound of Theorem 3. Let $\Delta(p) := P(p) - 8 + 4K/p$. Table V and Fig. 5(b) show $p^2 \Delta(p)$ approaching κ .

Fig. 5(a) illustrates the origin of K . Near the corner $(1, 1)$, put $x = 1 - \alpha/p$ and $y = 1 - \beta/p$. To leading order, the equation $x^p + y^p = 1$ becomes $e^{-\alpha} + e^{-\beta} = 1$, a single fixed curve that does not depend on p . The constant K is exactly the taxicab defect of this curve; parametrizing it by $e^{-\alpha} = t$ reproduces the integral for $F(0)$.



(a) The corner $(1, 1)$, magnified by the factor p .



(b) Rescaled perimeter defect $p(8 - P(p))$ against $1/p$.

Fig. 5. The superellipse. (a) Magnified by the factor p around the corner $(1, 1)$, the superellipse approaches the fixed curve $e^{-\alpha} + e^{-\beta} = 1$, which meets the sides of the square (thin lines) exponentially fast. (b) $p(8 - P(p))$ tends to $4K$ as $p \rightarrow \infty$, with slope close to $-\kappa$ as in (21).

TABLE V
PERIMETER OF THE SUPERELLIPSE $|x|^p + |y|^p = 1$

p	$P(p)$	$8 - 4K/p$	$p^2 \Delta(p)$
2	6.283185307180	5.786786801	1.986
4	7.017697943564	6.893393401	1.989
10	7.577408317258	7.557357360	2.005
20	7.783710054121	7.778678680	2.013
50	7.912278464092	7.911471472	2.017
100	7.955937655831	7.955735736	2.019
1000	7.995575594378	7.995573574	2.021

VIII. THE ORIGIN OF THE LOGARITHM

Formula (9) says that length is lost only while the curve moves in a direction strictly between horizontal and vertical. A steep piece of the curve, with slope $s \gg 1$, loses the fraction $g(s) \approx 1$ of its horizontal extent. The defect is therefore approximately the horizontal distance traveled while the curve is steep.

For $y = x^n$, the slope exceeds 1 exactly when $x \geq n^{-1/(n-1)}$, an interval of length

$$1 - n^{-1/(n-1)} = \frac{\ln n}{n} + O\left(\frac{\ln^2 n}{n^2}\right).$$

Near $x = 1$ we have $x^n \approx e^{-n(1-x)}$. To climb from height $1/n$, where the slope is about 1, to height 1, the curve must advance horizontally by about $\ln(n)/n$ (Fig. 6). The factor $\ln n$ is the logarithm of the ratio between the heights 1 and $1/n$ at which

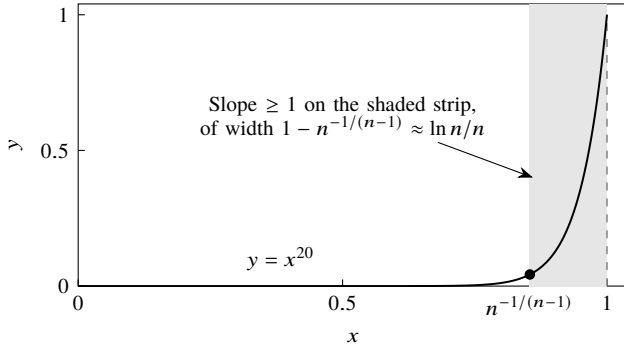


Fig. 6. Origin of the defect of x^n for $n = 20$. The curve has slope at least 1 only on the shaded strip. There it climbs from height about $1/n$ (dot) to 1 while moving horizontally by about $\ln n/n$, and almost all of that horizontal movement is lost relative to the corner path.

the steep part ends and begins. Equivalently, x^n approaches the vertical leg $x = 1$ only logarithmically, since at height y its distance to the leg is about $\ln(1/y)/n$.

The superellipse behaves differently. Its steep part is confined to a neighborhood of size $O(1/p)$ around the corner $(1, 1)$, where, as shown in Section VII, it looks like the fixed curve $e^{-\alpha} + e^{-\beta} = 1$ blown down by the factor $1/p$. That curve approaches its two asymptotes exponentially fast, so its total diagonal travel is finite, and the defect is K/p per corner.

To summarize, the defect is of order $1/n$ times the logarithm of the ratio between the largest and smallest heights at which the curve is steep. This ratio is bounded for the superellipse and is about n for x^n . The same reasoning accounts for the factor $1 - \cos \theta$ in Theorem 2, where only that fraction of each step is lost, and for the term $\ln c$ in Corollary 2, where stretching the vertical leg by c multiplies the ratio of heights by c .

IX. CONCLUSION AND OPEN PROBLEMS

When a family of curves collapses onto a corner, the length defect measures how far the curves travel in diagonal directions. For the graph of x^n , the defect is $(\ln n + 1 - \ln 2)/(n - 1)$ to first order. We derived this, and the complete asymptotic expansion, from an exact Gamma-function formula obtained from a single Beta integral. The logarithm persists for every turning angle, with coefficient $2 \sin^2(\theta/2)$, and it is absent for the superellipse, whose perimeter defect is $4K/p$. The difference lies in how fast each curve approaches the sides of the corner, which is logarithmic for x^n and exponential for the superellipse. We close with four open problems.

Problem 1: Is there a closed form for the second superellipse constant $\kappa = 2.02095276965 \dots$ in (21)? It equals $-4F'(0)$, an integral of logarithms against algebraic functions, and we expect dilogarithms to appear. An integer-relation search against π^2 , $\ln^2 2$, $\ln^2(1 + \sqrt{2})$, $\sqrt{2} \ln 2 \ln(1 + \sqrt{2})$, $\ln 2$, and $\sqrt{2} \ln(1 + \sqrt{2})$ found no relation.

Problem 2: Theorem 1 rests on the Mellin transform of $g = h_0$. Find the analogous exact formula for general angles, based on the Mellin transform of h_c in (11). We expect Legendre functions of $\cos \theta$ to appear.

Problem 3: The heuristic of Section VIII suggests the following. Let $f : [0, 1] \rightarrow [0, 1]$ be smooth and increasing, with $f(0) = 0$, $f(1) = 1$, and $f'(1) > 0$. Then the graph of f^n should satisfy $2 - \text{length}(f^n) \sim \ln n / (f'(1)n)$. Prove this and find the next term.

Problem 4: What happens in higher dimensions? For example, how does the area of the graph of $z = x^n y^n$ over $[0, 1]^2$ approach its limit?

APPENDIX A HIGHER-ORDER TERMS

Proposition 2: With $a_2 = \frac{\pi^2}{12} + \frac{1}{2}$ and $d = a_2(1 + \ln 2) + \frac{\zeta(3)}{4} - \frac{4}{3} = 1.2063122 \dots$, as $n \rightarrow \infty$,

$$L(n) = 1 + \left(\frac{2}{en}\right)^{\frac{1}{n-1}} + \frac{\pi^2}{12n^2} + \frac{d - a_2 \ln n}{n^3} + O\left(\frac{\ln^2 n}{n^4}\right).$$

Proof: Keep $\lambda = \ln n - a_1$ as in the proof of Corollary 1. By Lemma 3,

$$\frac{R(\varepsilon)}{n^\varepsilon} = e^{-\varepsilon\lambda} (1 + a_2 \varepsilon^2 + a_3 \varepsilon^3 + O(\varepsilon^4)).$$

Since $e^{-\varepsilon\lambda} = 1 - \varepsilon\lambda + O(\varepsilon^2 \ln^2 n)$, the product of $e^{-\varepsilon\lambda}$ with the bracket equals

$$e^{-\varepsilon\lambda} + a_2 \varepsilon^2 - a_2 \varepsilon^3 \lambda + a_3 \varepsilon^3 + O(\varepsilon^4 \ln^2 n).$$

The series in Theorem 1 equals $\frac{1}{2n(n-2)} + O(n^{-4}) = \frac{1}{2n^2} + \frac{1}{n^3} + O(n^{-4})$, and $\varepsilon^2 = \frac{1}{n^2} + \frac{2}{n^3} + O(n^{-4})$, $\varepsilon^3 = \frac{1}{n^3} + O(n^{-4})$. Collecting terms, the coefficient of n^{-2} is $a_2 - \frac{1}{2} = \frac{\pi^2}{12}$, and the coefficient of n^{-3} is

$$2a_2 - a_2(\ln n + 1 - \ln 2) + a_3 - 1 = d - a_2 \ln n,$$

because $a_3 - 1 = \frac{\zeta(3)}{4} - \frac{4}{3}$. ■

APPENDIX B NUMERICAL METHODS

All numerical values in this paper were computed in Python with the arbitrary-precision library mpmath, using 30 to 50 significant digits. The script `verify.py` in the accompanying repository recomputes every value and checks every claim listed below; all checks pass.

- *Arc lengths.* $L(n)$ was computed by tanh-sinh quadrature of (1), with breakpoints at $1 - k/n$ for $k \in \{0.05, 0.1, \dots, 50\}$, where the integrand varies rapidly. Theorem 1, summed with `nsum`, agrees with quadrature to better than 10^{-25} for $n \in \{2.5, 3, 5, 7.5, 10, 100, 1000\}$; so does (19) for $(c, n) \in \{(3, 5), (0.5, 7), (0.25, 9.5)\}$. Example 1 and Remark 1 were checked the same way.
- *Special functions.* Lemma 2 was checked at $\varepsilon = 0.37$ and 0.8 , using the substitution $s = e^u$ and the cancellation-free form $k(s) = 1/(s + \sqrt{1 + s^2})$. The Taylor coefficients in Lemma 3 were checked to 10^{-20} .
- *Asymptotics.* The bounds of Proposition 1 were checked for $3 \leq n \leq 10^5$, and Corollaries 1 and 2 and Proposition 2 up to $n = 10^5$.

- *Corners*. The constant (20) was compared with quadrature of the integrals defining C_{hc} , written in cancellation-free form, for five angles, to 10^{-20} . Table IV comes from quadrature of $L_\theta(n)$ at $n = 10^6$.
- *Superellipse*. The parametrization in the proof of Theorem 3 was checked against direct quadrature for $p \in \{2, 4, 10\}$, including $P(2) = 2\pi$, and the bound of Theorem 3 was checked for every p in Table V. The integer-relation search for κ used the PSLQ algorithm as implemented in `mpmath`.

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